

Lab 4: Normal-Normal Conjugacy, Posterior Predictive Distribution and Monte Carlo

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Part B. Computer analysis in R

1. Load and inspect the data.

```
library(gapminder)

d <- subset(gapminder, year == 2007)
d$log_gdp <- log(d$gdpPercap)
d$x <- d$log_gdp - mean(d$log_gdp)

summary(d[, c("lifeExp", "gdpPercap", "log_gdp")])
```

lifeExp	gdpPercap	log_gdp
Min. :39.61	Min. : 277.6	Min. : 5.626
1st Qu.:57.16	1st Qu.: 1624.8	1st Qu.: 7.393
Median :71.94	Median : 6124.4	Median : 8.720
Mean :67.01	Mean :11680.1	Mean : 8.616
3rd Qu.:76.41	3rd Qu.:18008.8	3rd Qu.: 9.799
Max. :82.60	Max. :49357.2	Max. :10.807

```
plot(d$log_gdp, d$lifeExp, pch = 19, col = "#2B78A6",
     xlab = "log(GDP per capita)", ylab = "Life expectancy (years)")
```

2. Fit the conjugate model. The next code block implements the normal-inverse-gamma posterior directly. We use weak but proper priors:

$$m_0 = (65, 0)^T, \quad V_0 = \text{diag}(100, 100), \quad a_0 = 2, \quad b_0 = 100.$$

```

y <- d$lifeExp
X <- cbind(1, d$x)
n <- nrow(X)

m0 <- c(65, 0)
V0 <- diag(c(100, 100))
a0 <- 2
b0 <- 100

Vn <- solve(solve(V0) + crossprod(X))
mn <- Vn %*% (solve(V0) %*% m0 + crossprod(X, y))
an <- a0 + n / 2
bn <- as.numeric(b0 + 0.5 * (
  crossprod(y) + t(m0) %*% solve(V0) %*% m0 -
  t(mn) %*% solve(Vn) %*% mn
))

mn

```

```

      [,1]
[1,] 67.007281
[2,]  7.202524

```

```

c(an = an, bn = bn)

```

```

      an      bn
73.000 3651.136

```

3. Draw from the joint posterior. Here, `1/rgamma(...)` gives inverse-gamma draws under the convention used in class.

```

set.seed(402)
S <- 10000
sigma2_draw <- 1 / rgamma(S, shape = an, rate = bn)

R <- chol(Vn)
beta_draw <- matrix(mn, nrow = S, ncol = 2, byrow = TRUE) +
  (matrix(rnorm(2 * S), nrow = S, ncol = 2) %*% R) *
  sqrt(sigma2_draw)

colnames(beta_draw) <- c("intercept", "slope")
apply(beta_draw, 2, quantile, probs = c(.025, .5, .975))

```

	intercept	slope
2.5%	65.81044	6.320846
50%	66.99885	7.208212
97.5%	68.19318	8.070102

```
mean(beta_draw[, "slope"] > 0)
```

```
[1] 1
```

4. Posterior and predictive summaries at \$20,000 GDP per capita.

```
g_star <- 20000
x_star <- log(g_star) - mean(d$log_gdp)
x_star_vec <- c(1, x_star)

mu_star_draw <- as.vector(beta_draw %*% x_star_vec)
y_star_draw <- rnorm(S, mean = mu_star_draw, sd = sqrt(sigma2_draw))

quantile(mu_star_draw, c(.025, .5, .975))
```

2.5%	50%	97.5%
74.68113	76.28301	77.87913

```
quantile(y_star_draw, c(.025, .5, .975))
```

2.5%	50%	97.5%
61.93719	76.21110	90.94310

Part C. What should you report?

1. The 95% credible interval for the slope is (6.32, 8.07). This says that there is a 95% probability that for every 1 unit increase in log GDP per capita, the expected life expectancy increases by 6.32 - 8.07 years.
2. Estimate $Pr(\beta_1 > 0 \mid y)$ from the posterior draws. What does this posterior probability summarize?

```
mean(beta_draw[, "slope"] > 0)
```

[1] 1

This posterior probability summarizes how certain we are that the slope is positive, which, in our case, exceeds 0.9999.

1. Report a 95% credible interval for the conditional mean life expectancy at GDP per capita equal to \$20,000. Then report a 95% predictive interval for one new country's life expectancy at the same GDP level. Why are they different?
2. Make a plot of the data with the posterior mean regression line: